

JEE MAINS QFT-17 2025



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By: C I P H E R

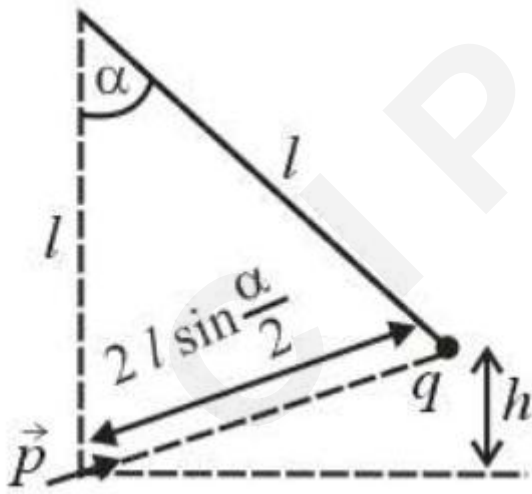


Question Paper

Physics Single Correct (Maximum Marks: 80)

Question No. 1 (Only One Option Correct Type)

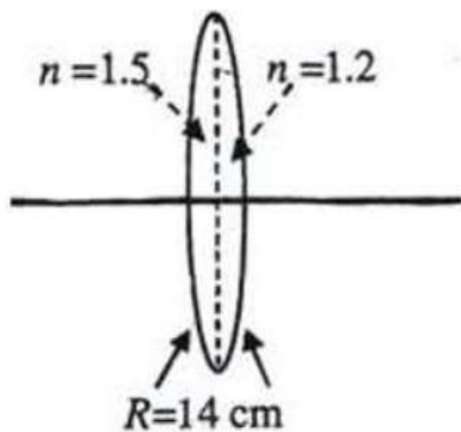
A point charge q of mass m is suspended vertically by a string of length l . A point dipole of dipole moment $\vec{p}$ is now brought towards q from infinity so that the charge moves away. The final equilibrium position of the system including the direction of the dipole, the angles and distances is shown in the figure below. If the work done in bringing the dipole to this position is $N \times (mgh)$, where g is the acceleration due to gravity, then the value of N is _____ (Note that for three coplanar forces keeping a point mass in equilibrium, $\frac{F}{\sin\theta}$ is the same for all forces, where F is any one of the forces and θ is the angle between the other two forces)



- (A) 4
- (B) 2
- (C) 3
- (D) 5

Question No. 2 (Only One Option Correct Type)

A bi-convex lens is formed with two thin plano-convex lenses as shown in the figure. Refractive index n of the first lens is 1.5 and that of the second lens is 1.2. Both the curved surfaces are of the same radius of curvature $R = 14$ cm. for this bi-convex lens, for an object distance of 40 cm, the image distance will be



- (A) -280.0 cm
- (B) 40.0 cm
- (C) 21.5 cm
- (D) 13.3 cm

Question No. 3 (Only One Option Correct Type)

Which statement(s) is/are true?

- (i) Kirchhoff's law is equally applicable to both AC and DC.
- (ii) Semiconductors have a positive temperature coefficient of resistance.
- (iii) Meter bridge has a greater sensitivity when the resistances of all the four arms of the bridge are of the same order.
- (iv) The emf of a cell depends upon the size and area of electrodes.

- A. (i) and (iv)
- B. (ii) and (iv)
- C. (iii) and (iv)
- D. None of these

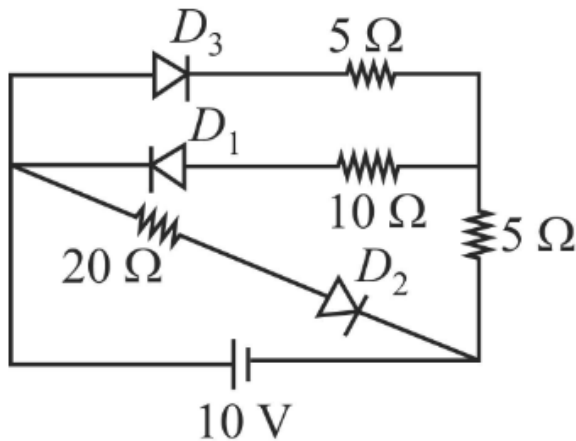
Question No. 4 (Only One Option Correct Type)

Two thin biconvex lenses have focal lengths f_1 and f_2 . A third thin biconcave lens has focal length of f_3 . If the two biconvex lenses are in contact, then the total power of the lenses is P_1 . If the first convex lens is in contact with the third lens, then the total power is P_2 . If the second lens is in contact with the third lens, the total power is P_3 , then

- A. $P_1 = \frac{f_1 f_2}{f_1 - f_2}$, $P_2 = \frac{f_1 f_3}{f_3 - f_1}$ and $P_3 = \frac{f_2 f_3}{f_3 - f_2}$
- B. $P_1 = \frac{f_1 - f_2}{f_1 f_2}$, $P_2 = \frac{f_3 - f_1}{f_3 + f_1}$ and $P_3 = \frac{f_3 - f_2}{f_2 f_3}$
- C. $P_1 = \frac{f_1 - f_2}{f_1 f_2}$, $P_2 = \frac{f_3 - f_1}{f_1 f_3}$ and $P_3 = \frac{f_3 - f_2}{f_2 f_3}$
- D. $P_1 = \frac{f_1 + f_2}{f_1 f_2}$, $P_2 = \frac{f_3 - f_1}{f_1 f_3}$ and $P_3 = \frac{f_3 - f_2}{f_2 f_3}$

Question No. 5 (Only One Option Correct Type)

In the given circuit, The current through battery is



A. 0.5 A

B. 1 A

C. 1.5 A

D. 2.5 A

Question No. 6 (Only One Option Correct Type)

Assertion: Small water drops are spherical while bigger water drops are not.

Reason: In small water drops, surface tension forces dominate, while in bigger water drops, gravity forces dominate.

A) Both Assertion and Reason are correct and Reason is the correct explanation for Assertion.

B) Both Assertion and Reason are correct but Reason is not the correct explanation for Assertion.

C) Assertion is correct but Reason is incorrect.

D) Assertion is incorrect but the reason is correct.

Question No. 7 (Only One Option Correct Type)

A circular hole of radius 1 cm is cut off from a disc of radius 6 cm. The centre of the hole is 3 cm from the centre of the disc. Then the distance of the centre of mass of the remaining disc from the centre of the disc is

A. $\frac{3}{35}$ cm B. $\frac{1}{35}$ cm C. $\frac{3}{10}$ cm D. None of these

Question No. 8 (Only One Option Correct Type)

The Power (P) radiated from an accelerated charged particle is given by

$P \propto \frac{1}{\epsilon_0} \frac{(qa)^m}{c^n}$ where q is the charge, a is the acceleration of the particle and c is speed of light in vacuum. From dimensional analysis, the value of m and n respectively, are

A. $m = 2, n = 2$

B. $m = 2, n = 3$

C. $m = 3, n = 2$

D. $m = 0, n = 1$

Question No. 9 (Only One Option Correct Type)

In a photoelectric experiment, with light of wavelength λ , the fastest electron has speed v . If the exciting wavelength is changed to $\frac{3\lambda}{4}$, the speed of the fastest emitted electron will become:

A. $v\sqrt{\frac{3}{4}}$

B. $v\sqrt{\frac{4}{3}}$

C. less than $v\sqrt{\frac{3}{4}}$

D. greater than $v\sqrt{\frac{4}{3}}$

Question No. 10 (Only One Option Correct Type)

To increase the current sensitivity of a moving coil galvanometer by 50%, its resistance is increased so that the new resistance becomes twice its initial resistance. By what factor does its voltage sensitivity change?

A. Increases by 15%

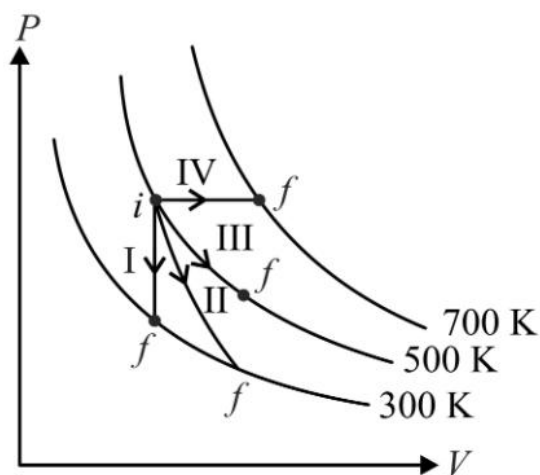
B. Decreases by 15%

C. Increases by 25%

D. Decreases by 25%

Question No. 11 (Only One Option Correct Type)

Thermodynamic processes are indicated in the following diagram.



Match the following

Column - 1	Column - 2
P. Process I	a. Adiabatic
Q. Process II	b. Isobaric
R. Process III	c. Isochoric
S. Process IV	d. Isothermal

A. $P \rightarrow a, Q \rightarrow c, R \rightarrow d, S \rightarrow b$

B. $P \rightarrow c, Q \rightarrow a, R \rightarrow d, S \rightarrow b$

C. $P \rightarrow c, Q \rightarrow d, R \rightarrow b, S \rightarrow a$

D. $P \rightarrow d, Q \rightarrow b, R \rightarrow a, S \rightarrow c$

Question No. 12 (Only One Option Correct Type)

A plane electromagnetic wave traveling along the x -direction has a wavelength of 3 mm. The variation in the electric field occurs in the y -direction with an amplitude of 66 V/m. The equations for the electric and magnetic fields as a function of x and t are respectively:

A. $E_y = 33 \cos \left(\pi \times 10^{11} \left(t - \frac{x}{c} \right) \right)$ and $B_z = 1.1 \times 10^{-7} \cos \left(\pi \times 10^{11} \left(t - \frac{x}{c} \right) \right)$

B. $E_y = 11 \cos \left(2\pi \times 10^{11} \left(t - \frac{x}{c} \right) \right)$ and $B_y = 11 \times 10^{-7} \cos \left(2\pi \times 10^{11} \left(t - \frac{x}{c} \right) \right)$

C. $E_x = 33 \cos \left(\pi \times 10^{11} \left(t - \frac{x}{c} \right) \right)$ and $B_y = 11 \times 10^{-7} \cos \left(\pi \times 10^{11} \left(t - \frac{x}{c} \right) \right)$

D. $E_y = 66 \cos \left(2\pi \times 10^{11} \left(t - \frac{x}{c} \right) \right)$ and $B_z = 2.2 \times 10^{-7} \cos \left(2\pi \times 10^{11} \left(t - \frac{x}{c} \right) \right)$

Question No. 13 (Only One Option Correct Type)

An ice block at 0°C is dropped from a height h above the ground. What should be the value of h (approximately) so that it just melts completely by the time it reaches the bottom, assuming the loss of the whole gravitational potential energy is used as heat by the ice? [Given, $L_f = 80 \text{ cal/g}$]:

A. 33.6 m

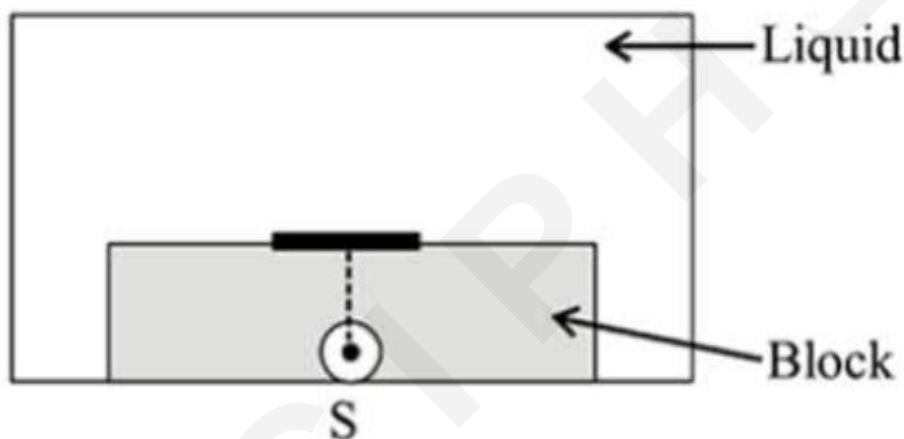
B. 33.6 km

C. 8 m

D. 8 km

Question No. 14 (Only One Option Correct Type)

A point source S is placed at the bottom of a transparent block of height 10 mm and refractive index 2.72. It is immersed in a lower refractive index liquid as shown in the figure. It is found that the light emerging from the block to the liquid forms a circular bright spot of diameter 11.54 mm on the top of the block. The refractive index of the liquid is:



A. 1.21

B. 1.3

C. 1.36

D. 1.42

Question No. 15 (Only One Option Correct Type)

The half-life of a radioactive substance A is two times the half-life of another radioactive substance B . Initially, the number of nuclei of A and B are N_A and N_B respectively. After three half-lives of A , the number of nuclei of both are equal. Then the ratio $\frac{N_A}{N_B}$ is:

- A. $\frac{1}{4}$
- B. $\frac{1}{8}$
- C. $\frac{1}{3}$
- D. $\frac{1}{6}$

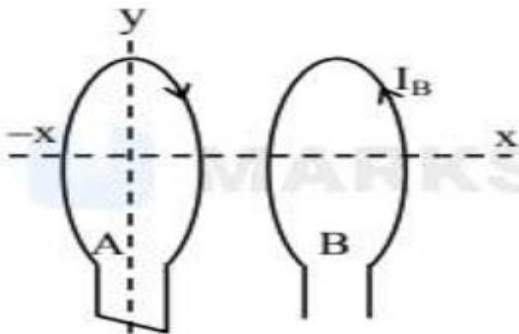
Question No. 16 (Only One Option Correct Type)

A piece of wood has length, breadth, and height a , b , and c respectively. Its relative density is d . It is floating in water such that the side a is vertical. It is pushed down a little and released. The time period of the S.H.M. executed by it is (g = acceleration due to gravity):

- A. $2\pi\sqrt{\frac{abc}{g}}$
- B. $2\pi\sqrt{\frac{bc}{dg}}$
- C. $2\pi\sqrt{\frac{g}{da}}$
- D. $2\pi\sqrt{\frac{ad}{g}}$

Question No. 17 (Only One Option Correct Type)

The diagram below shows two circular loops of wire (A and B) centered on and perpendicular to the x -axis, and oriented with their planes parallel to each other. The y -axis passes vertically through loop A (dashed line). There is a current I_B in loop B as shown. Possible actions which we might perform on loop A are:



- (i) Move A to the right along x axis closer to B
 - (ii) Move A to the left along x axis away from B
 - (iii) As viewed from above, rotate A clockwise about y axis
 - (iv) As viewed from above, rotate A anticlockwise about y axis
- Which of these actions will induce a current in A only in the direction shown.

A. Only (i)

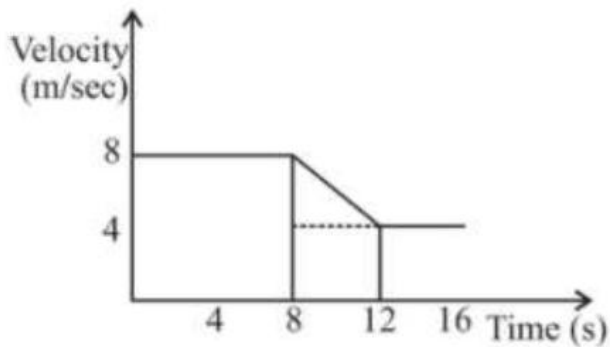
B. Only (ii)

C. Only (i) and (iv)

D. Only (ii) and (iii)

Question No. 18 (Only One Option Correct Type)

Consider the following velocity and time graph.



Which one of the following is the value of average acceleration from 8 seconds to 12 seconds?

A. 8 m/s^2

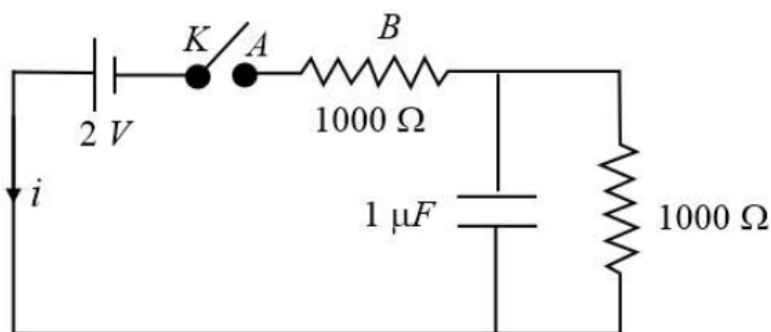
B. 12 m/s^2

C. 2 m/s^2

D. -1 m/s^2

Question No. 19 (Only One Option Correct Type)

When the key is pressed at time $t = 0$, then which of the following statements about current i in the resistor AB of the given circuit is true?



A. At $t = 0$, $i = 2 \text{ mA}$ and with time, it goes to 1 mA .

B. i oscillates between 1 mA and 2 mA .

C. $i = 2 \text{ mA}$ at all t .

D. $i = 1 \text{ mA}$ at all t .

Question No. 20 (Only One Option Correct Type)

A solid cylinder is released from rest from the top of an inclined plane of inclination 30° and length 60 cm . If the cylinder rolls without slipping, then the speed when it reaches the bottom is:

A. 3 m/s

B. 2.5 m/s

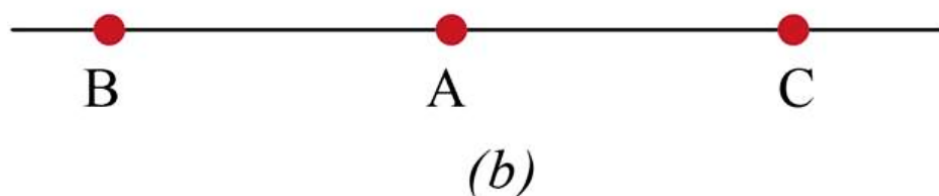
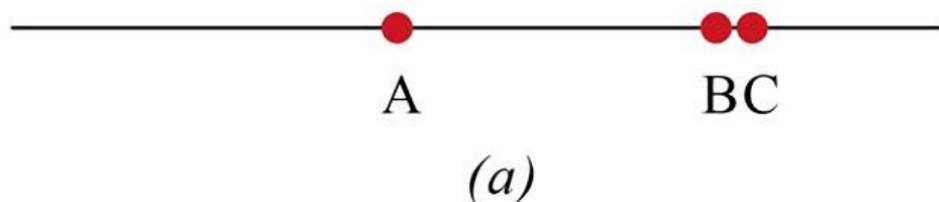
C. 2 m/s

D. 3.5 m/s

Physics Numerical (Maximum Marks: 20)

Question No. 1 (Numerical Type)

In figure (a), three positively charged particles are fixed on an x -axis. Particles B and C are so close to each other that they can be considered to be at the same distance from particle A . The net force on particle A due to particles B and C is $F \text{ N}$ in the negative direction of the x -axis. In figure (b), particle B has been moved to the opposite side of A but is still at the same distance from it. The net force on A is now $\frac{F}{7} \text{ N}$ in the negative direction of the x -axis. The ratio of the charge of particle C to that of particle B is $4 : b$. Then the value of b is:



Question No. 2 (Numerical Type)

Current through an AC series circuit is 4 A, if operated at resonant frequency ω_0 . At $\frac{\omega_0}{2}$, current reduces to 2 A. Then, at $2\omega_0$, the current in the circuit shall be:

Question No. 3 (Numerical Type)

In a two-dimensional motion of a particle, the particle moves from point A , position vector $\mathbf{r}_1$, to point B , position vector $\mathbf{r}_2$. If the magnitudes of these vectors are respectively $r_1 = 3$ and $r_2 = 4$, and the angles they make with the x -axis are $\theta_1 = 75^\circ$ and $\theta_2 = 15^\circ$ respectively, then find the magnitude of the displacement vector. If it is $\sqrt{n} + 4$, find n .

Question No. 4 (Numerical Type)

During an adiabatic expansion, a gas does 50 J of work against the surroundings. It is then cooled at constant volume by removing 20 J of energy from the gas. The magnitude of the total change in internal energy of the gas is x J. The value of x is:

Question No. 5 (Numerical Type)

A particle moves along the X -axis from $x = 0$ to $x = 5$ m under the influence of a force given by

$$F = (7 - 2x + 3x^2) \text{ N}.$$

The work done in the process is (Answer in J):

Chemistry Single Correct (Maximum Marks: 80)

Question No. 1 (Only One Option Correct Type)

Statement I: A unit formed by the attachment of a base to 1' position of sugar is known as nucleoside

Statement II: When nucleoside is linked to phosphorous acid at 5'-position of sugar moiety, we get nucleotide.

In the light of the above statements, choose the correct answer from the options given below:

- A. Statement I is false but Statement II is true.
- B. Both Statement I and Statement II are true.
- C. Both Statement I and Statement II are false.
- D. Statement I is true but Statement II is false.

Question No. 2 (Only One Option Correct Type)

Which of the following complex will show fac & mer isomerism?

- A. $[\text{Co}(\text{NH}_3)_3(\text{NO}_2)_3]$
- B. $[\text{CoCl}_2(\text{en})_2]^+$
- C. $[\text{Co}(\text{NH}_3)\text{Cl}(\text{en})_2]^{2+}$
- D. $[\text{Co}(\text{NH}_3)_2\text{Cl}_2(\text{en})]^+$

Question No. 3 (Only One Option Correct Type)

Given E° for $\text{Cu}^{2+} \rightarrow \text{Cu}^+$ is $+0.15 \text{ V}$ and $\text{Cu}^+ \rightarrow \text{Cu}$ is $+0.50 \text{ V}$. Calculate E° for $\text{Cu}^{2+} \rightarrow \text{Cu}$ (Round off 10 times E to the nearest integer as answer):

- A. 2
- B. 3
- C. 4
- D. 1

Question No. 4 (Only One Option Correct Type)

Match List-I with List-II :

	List-I (Atom/Molecule)		List-II (Property)
A.	Nitrogen atom	I.	Paramagnetic
B.	Fluorine molecule	II.	Most reactive element in group 18
C.	Oxygen molecule	III.	Element with highest ionisation enthalpy in group 15
D.	Xenon atom	IV.	Strongest oxidising agent

A. A-III, B-I, C-IV, D-II

B. A-I, B-IV, C-III, D-II

C. A-II, B-IV, C-I, D-III

D. A-III, B-IV, C-I, D-II

Question No. 5 (Only One Option Correct Type)

Select the correct option by assigning true (T) and false (F) for the following statements:

(a) In IF_7 , iodine is in the super octet state.

(b) BF_3 is a hypervalent compound.

(c) π -bond is cylindrically symmetrical.

A. TFF

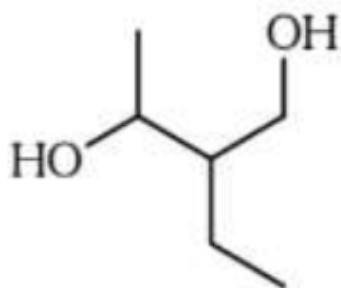
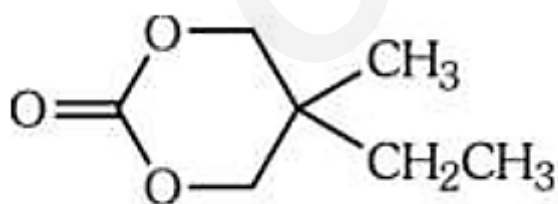
B. FFF

C. TTT

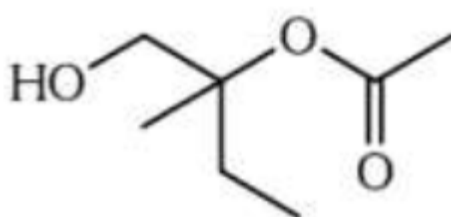
D. FTF

Question No. 6 (Only One Option Correct Type)

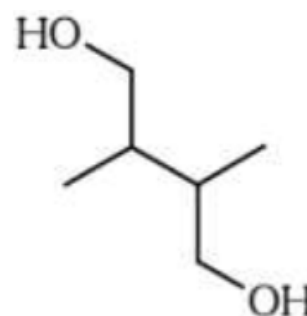
On being heated with a dilute solution of sulfuric acid in water, the compound shown was observed to liberate CO_2 . What organic compound is formed during the reaction?



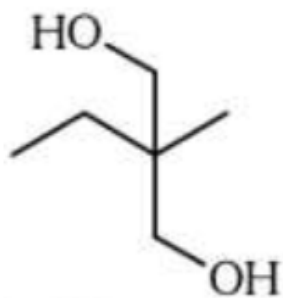
(A)



(B)



(C)



(D)

Question No. 7 (Only One Option Correct Type)

In order to distinguish between $\text{C}_2\text{H}_5\text{NH}_2$ and $\text{C}_6\text{H}_5\text{NH}_2$, which of the following reagents is useful?

A. Hinsberg reagent

B. β -Naphthol

C. Benzene diazonium chloride

D. None of these

Question No. 8 (Only One Option Correct Type)

Match List I and List II and choose the correct answer by the codes given below the lists.

List I	List II
A. Uranium oxide	1. Red
B. Cuprous oxide	2. Blue
C. Cobalt oxide	3. Green
D. Chromium oxide	4. Yellow

A. A-4, B-1, C-2, D-3

B. A-3, B-2, C-1, D-4

C. A-4, B-2, C-1, D-3

D. A-3, B-1, C-2, D-4

Question No. 9 (Only One Option Correct Type)

In the Lassaigne's test for the detection of nitrogen in an organic compound, the appearance of a blue-colored compound is due to:

A. ferric ferricyanide

B. ferrous ferricyanide

C. ferric ferrocyanide

D. ferrous ferrocyanide

Question No. 10 (Only One Option Correct Type)

How many dichloro products (including stereoisomers) will be formed when R – 2 – chloropentane reacts with Cl_2 in the presence of UV radiation?

A. 7

B. 6

C. 5

D. 8

Question No. 11 (Only One Option Correct Type)

What is the correct order of magnetic moments (spin values in BM) among $[\text{MnCl}_4]^{2-}$, $[\text{Fe}(\text{CN})_6]^{4-}$, and $[\text{CoCl}_4]^{2-}$?

(Atomic number: Mn = 25, Fe = 26, Co = 27)

A. $[\text{Fe}(\text{CN})_6]^{4-} > [\text{CoCl}_4]^{2-} > [\text{MnCl}_4]^{2-}$

B. $[\text{MnCl}_4]^{2-} > [\text{CoCl}_4]^{2-} > [\text{Fe}(\text{CN})_6]^{4-}$

C. $[\text{Fe}(\text{CN})_6]^{4-} > [\text{MnCl}_4]^{2-} > [\text{CoCl}_4]^{2-}$

D. $[\text{MnCl}_4]^{2-} > [\text{Fe}(\text{CN})_6]^{4-} > [\text{CoCl}_4]^{2-}$

Question No. 12 (Only One Option Correct Type)

What will be the result if 100 mL of 0.06 M $\text{Mg}(\text{NO}_3)_2$ solution is added to 50 mL of 0.06 M $\text{Na}_2\text{C}_2\text{O}_4$ solution? The K_{sp} of MgC_2O_4 (s) is $8.6 \times 10^{-5} \text{ M}^2$.

A. No precipitate will form.

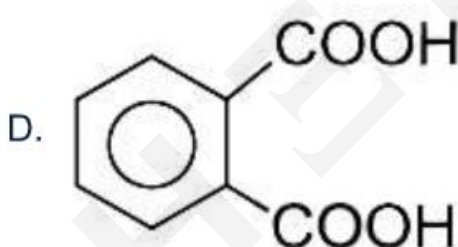
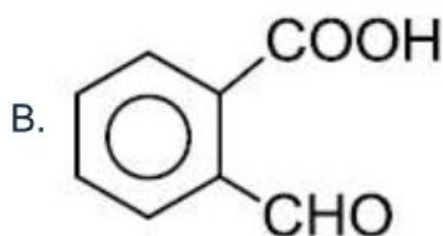
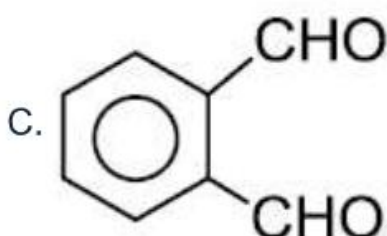
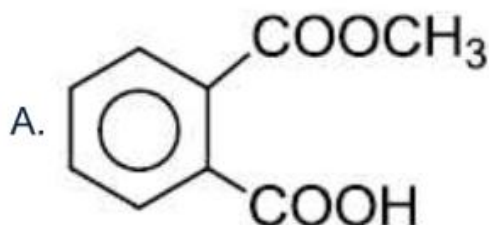
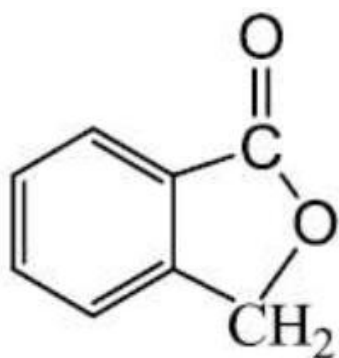
B. A precipitate will form and excess of Mg^{2+} will remain in solution.

C. A precipitate will form and excess of $\text{C}_2\text{O}_4^{2-}$ ions will remain in solution.

D. A precipitate will form but neither Mg^{2+} ions nor $\text{C}_2\text{O}_4^{2-}$ ions will remain in solution.

Question No. 13 (Only One Option Correct Type)

Which of the following reactants, on reaction with concentrated NaOH followed by acidification, gives the following lactone or cyclic ester as the product?



Question No. 14 (Only One Option Correct Type)

What will be the molar entropy change when silver is heated from 227°C to 727°C ? Consider the molar specific heat of Ag, $C_p = 5.3 + 0.0028 T$:

A. 7.873 eu

B. 12.341 eu

C. 3.214 eu

D. 5.074 eu

Question No. 15 (Only One Option Correct Type)

Identify the incorrect statement:

A. PEt_3 and AsPh_3 as ligands can form $d\pi - d\pi$ bond with transition metals.

B. The N-N single bond is as strong as the P-P single bond.

C. Nitrogen has a unique ability to form $p\pi - p\pi$ multiple bonds with nitrogen, carbon, and oxygen.

D. Nitrogen cannot form $d\pi - p\pi$ bond as other heavier elements of its group.

Question No. 16 (Only One Option Correct Type)

Mr. Parker consumes 1 kg of bread on average every day. Bread contains 10% by mass of protein. Protein contains 7% by mass of nitrogen. If 10% of the nitrogen present in protein is converted to urea, how much urea is produced in Mr. Parker in one day?

- A. 750 mg
- B. 70 mg
- C. 1500 mg
- D. 150 mg

Question No. 17 (Only One Option Correct Type)

Match the following.

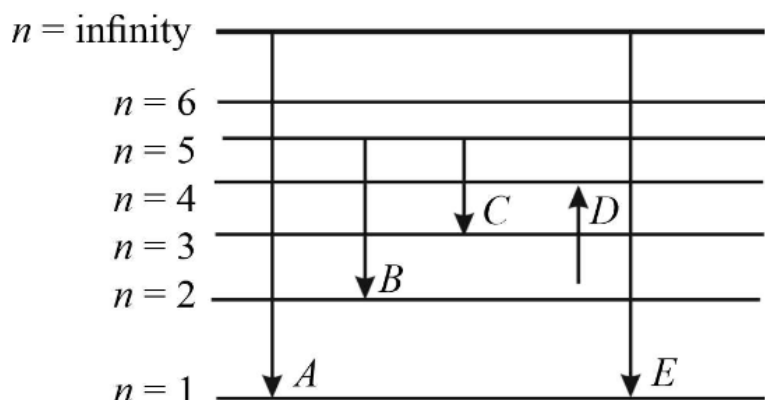
List I	List II
A. The reaction of 1, 6-dibromo hexane with Zn.	i. $\text{H}_3\text{C}-\text{C}\equiv\text{CH}$
B. Reaction of ethanol with concentrated H_2SO_4 at 443 K.	ii. $\text{H}_2\text{C}=\text{CH}_2$
C. Major product in the reaction of propene with HBr in the presence of benzoyl peroxide.	iii. 
D. The reaction of 1, 1-dibromopropane with NaNH_2 at 433 K.	iv. 

The correct answer is

- A. A-iv, B-ii, C-iii, D-i
- B. A-iii, B-i, C-ii, D-iv
- C. A-ii, B-iii, C-i, D-iv
- D. A-i, B-ii, C-iv, D-iii

Question No. 18 (Only One Option Correct Type)

The energy levels of the hydrogen spectrum is shown in the figure. There are some transitions A, B, C, D and E . Transition A, B and C respectively represent



First member of Lyman series, third spectral line of Balmer series and the
A. second spectral line of Paschen series.

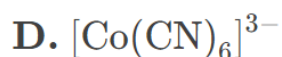
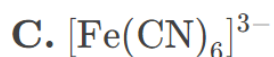
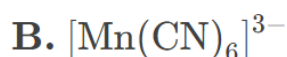
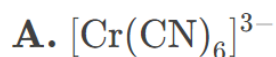
Ionisation potential of hydrogen, second spectral line of Balmer series and third
B. spectral line of Paschen series.

Series limit of Lyman series, third spectral line of Balmer series and second
C. spectral line of Paschen series.

Series limit of Lyman series, second spectral line of Balmer series and third
D. spectral line of Paschen series.

Question No. 19 (Only One Option Correct Type)

Which one of the following coordination complexes exhibits the lowest value of magnetic moment (in BM)?



Question No. 20 (Only One Option Correct Type)

Assertion (A): The $-\text{NH}_2$ group of aniline is ortho, para-directing in electrophilic substitutions.

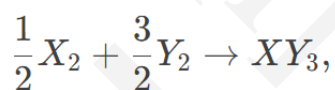
Reason (R): The $-\text{NH}_2$ group stabilizes the arenium ion formed by the ortho, para-attack of the electrophile.

- A. Both (A) and (R) are correct, and (R) is the correct explanation of (A).
- B. Both (A) and (R) are correct, but (R) is not the correct explanation of (A).
- C. (A) is correct, but (R) is not correct.
- D. (A) is not correct, but (R) is correct.

Chemistry Numerical (Maximum Marks: 20)

Question No. 1 (Numerical Type)

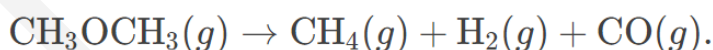
The standard entropy of X_2Y_2 and XY_3 are 60, 40, and $50 \text{ JK}^{-1}\text{mol}^{-1}$, respectively. For the reaction



$\Delta H = -30 \text{ kJ/mol}$. To be at equilibrium, the temperature will be (Mark answer in Kelvin and round off to the nearest integer):

Question No. 2 (Numerical Type)

The gas phase decomposition of dimethylether follows first-order kinetics:



The reaction is carried out in a constant volume container at 500°C and has a half-life of 14.5 minutes. Initially, only dimethylether is present at a pressure of 0.40 atm. The total pressure of the system after 12 minutes is $\frac{x}{100}$ atm. The value of x is:

$$[\text{Given } 10^{0.25} = 1.778]$$

Question No. 3 (Numerical Type)

In Duma's method for the estimation of nitrogen, 0.25 g of an organic compound gave 40 mL of nitrogen collected at 300 K temperature and 725 mm pressure. If the aqueous tension at 300 K is 25 mm, the percentage of nitrogen in the compound is: (Nearest Integer)

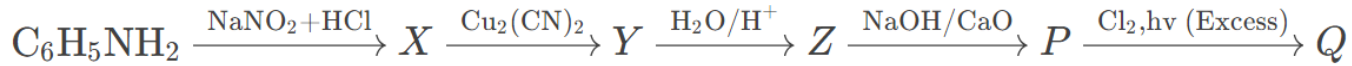
Question No. 4 (Numerical Type)

A 100 mL, 0.1 M solution of ammonium acetate is diluted by adding 100 mL of water. The pH of the resulting solution will be:

($\text{p}K_a$ of acetic acid is nearly equal to $\text{p}K_b$ of NH_4OH)

Question No. 5 (Numerical Type)

Identify the total number of atoms in product Q in the given sequence of reactions:



Mathematics Single Correct (Maximum Marks: 80)

Question No. 1 (Only One Option Correct Type)

Let a, b, c, d be four distinct real numbers in A.P. Find the smallest positive value of k satisfying

$$2(a - b) + k(b - c)^2 + (c - a)^3 = 2(a - d) + (b - d)^2 + (c - d)^3$$

A. 18 B. 20 C. 16 D. 15

Question No. 2 (Only One Option Correct Type)

Let a red die, a blue die, a green die, and a white die be rolled once, the dice being fair. The outcomes on the red, blue, green, and white die denote the numbers a, b, c , and d respectively. Let E denote the event that the absolute value of

$$(a - 1)(b - 2)(c - 3)(d - 6) = 1$$

then $P(E)$ is:

A. $\frac{1}{324}$ B. $\frac{1}{648}$ C. $\frac{2}{324}$ D. $\frac{1}{160}$

Question No. 3 (Only One Option Correct Type)

Let O be an interior point of $\triangle ABC$, such that

$$\overrightarrow{OA} + 2\overrightarrow{OB} + 3\overrightarrow{OC} = \mathbf{0}$$

Then the ratio of the area of $\triangle ABC$ to the area of $\triangle AOC$ is:

A. 2 B. $\frac{3}{2}$ C. 3 D. $\frac{5}{2}$

Question No. 4 (Only One Option Correct Type)

Let

$$f(x) = \begin{cases} x, & x < 0 \\ 1 + x^2, & x \geq 0 \end{cases}$$

and

$$g(x) = 1 + x - [x]$$

where $[.]$ denotes the greatest integer function. Then the range of $f \circ g(x)$ is:

A. $[1, 5]$ **B.** $[2, 5]$ **C.** $(2, 5]$ **D.** $[1, 5]$

Question No. 5 (Only One Option Correct Type)

If

$$A^{-1} = \begin{bmatrix} \sin^2 \alpha & 0 & 0 \\ 0 & \sin^2 \beta & 0 \\ 0 & 0 & \sin^2 \gamma \end{bmatrix}$$

and

$$B^{-1} = \begin{bmatrix} \cos^2 \alpha & 0 & 0 \\ 0 & \cos^2 \beta & 0 \\ 0 & 0 & \cos^2 \gamma \end{bmatrix}$$

where $\alpha, \beta, \gamma \in \mathbb{R}$ and

$$C = A^{-5} + B^{-5} + 5A^{-1}B^{-1}(A^{-3} + B^{-3}) + 10A^{-2}B^{-2}(A^{-1} + B^{-1})$$

then $C =$

A. I **B.** $5I$ **C.** $32I$ **D.** $2(A^{-1} + B^{-1})$

Question No. 6 (Only One Option Correct Type)

For $2n$ observations $a_1, -a_1, a_2, -a_2, a_3, -a_3, \dots, a_n, -a_n$ where all a 's are distinct. Let $\bar{x}, \sigma$ and M denote the mean, standard deviation, and median, respectively. Then which of the following options is incorrect when n is an odd integer?

A. $\bar{x} = 0$ **B.** $\bar{x} = M$ **C.** $\sigma > \bar{x}$ **D.** $\sigma < M$

Question No. 7 (Only One Option Correct Type)

If the system of linear equations

$$2x + y - z = 7$$

$$x - 3y + 2z = 1$$

$$x + 4y + \delta z = k$$

where $\delta, k \in \mathbb{R}$ has infinitely many solutions, then $\delta + k$ is equal to:

A. -3 B. 3 C. 6 D. 9

Question No. 8 (Only One Option Correct Type)

Triangle ABC lies in the Cartesian plane and has an area of 70 sq. units. The coordinates of B and C are $(12, 19)$ and $(23, 20)$ respectively, and the coordinates of A are (p, q) . The median to the side BC has a slope of -5 . Then which of the following can be correct?

A. $p + q = 46$ B. $p + q = 26$ C. $p - q = 13$ D. $p - q = 17$

Question No. 9 (Only One Option Correct Type)

The solution of the differential equation

$$\frac{dy}{dx} + \frac{xy}{1 - x^2} = x\sqrt{y}, \quad (|x| < 1)$$

is

$$\sqrt{y} = -\frac{f(x)}{3} + C(1 - x^2)^{\frac{1}{4}},$$

where $f\left(\frac{1}{2}\right) = \frac{3}{4}$ and C is an arbitrary constant. Then, the value of $f\left(-\frac{1}{2}\right)$ is:

A. $-\frac{3}{4}$ B. $\frac{3}{4}$ C. $\frac{1}{4}$ D. $\frac{3}{2}$

Question No. 10 (Only One Option Correct Type)

Consider a group of 5 boys and 7 girls. The number of different teams, consisting of 2 boys and 3 girls that can be formed from this group if there are two specific girls A and B, who refuse to be the members of the same team, is....

A. 350 B. 300 C. 200 D. 500

Question No. 11 (Only One Option Correct Type)

For $f(x) = x^2 - 2|x|$, test the continuity and differentiability of $g(x)$ in the interval $[-2, 3]$, where

$$g(x) = \begin{cases} \min\{f(t); -2 \leq t \leq x\}, & -2 \leq x < 0 \\ \max\{f(t); 0 \leq t \leq x\}, & 0 \leq x \leq 3 \end{cases}$$

A. $g(x)$ is continuous and differentiable everywhere except $x = 0$

B. $g(x)$ is non-differentiable at $x = 0$ and 2

C. $g(x)$ is non-differentiable at $x = 0, 1, 2$

D. None of the above

Question No. 12 (Only One Option Correct Type)

A variable parabola having the length of its latus rectum equal to $4a$ and its axis parallel to the x -axis is drawn in such a way that it always touches the parabola $y^2 = 4ax$. The locus of the vertex of the variable parabola is:

A. $y^2 = 8ax$ **B.** $y^2 + 8ax = 0$ **C.** $y^2 = 6ax$ **D.** $y^2 + 6ax = 0$

Question No. 13 (Only One Option Correct Type)

A line with direction ratios $(2, 1, 2)$ intersects the lines

$$\mathbf{r} = -\mathbf{j} + \lambda(\mathbf{i} + \mathbf{j} + \mathbf{k})$$

and

$$\mathbf{r}' = -\mathbf{i} + \mu(2\mathbf{i} + \mathbf{j} + \mathbf{k})$$

at A and B . If the shortest distance of the origin from the line AB is d , then the value of $27d^2$ is:

A. 3 **B.** 6 **C.** 4 **D.** 8

Question No. 14 (Only One Option Correct Type)

The circles

$$x^2 + y^2 = a^2$$

and

$$(x - 2a)^2 + y^2 = a^2$$

are two equal circles touching each other. Find the equation of the circle (or circles) of the same radius touching both the circles.

A. $x^2 + y^2 + 2ax + 2\sqrt{3}ay + 3a^2 = 0$

B. $x^2 + y^2 - 2ax + 2\sqrt{3}ay + 3a^2 = 0$

C. $x^2 + y^2 + 2ax - 2\sqrt{3}ay + 3a^2 = 0$

D. None of these

Question No. 15 (Only One Option Correct Type)

The value of

$$I = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{x^2 \cos x}{1 + e^{-x}} dx$$

is equal to:

A. $\frac{\pi^2}{4} - 2$ **B.** $\frac{\pi^2}{4} + 2$ **C.** $\pi^2 - e^{\frac{\pi}{2}}$ **D.** $\pi^2 + e^{\frac{\pi}{2}}$

Question No. 16 (Only One Option Correct Type)

Let R be a relation defined as $R = \{(1, 2), (2, 3), (3, 4)\}$. The minimum number of ordered pairs that should be added to make R an equivalence relation is:

A. 7 **B.** 9 **C.** 13 **D.** 11

Question No. 17 (Only One Option Correct Type)

Let A be the region of the complex plane

$$\{z : z/4 \text{ and } 4/z \text{ have real and imaginary parts in } (0, 1)\}$$

Then $[p]$ (where p is the area of the region A and $[.]$ denotes the greatest integer function) is:

A. 25 **B.** 0 **C.** 5 **D.** 6

Question No. 18 (Only One Option Correct Type)

The expression

$$\frac{2 \tan A}{1 - \cot A} + \frac{2 \cot A}{1 - \tan A}$$

can be written as:

A. $\sin 2A + \cos 2A$ **B.** $2 \sec A \csc A + 2$ **C.** $\tan 2A + \cot 2A$ **D.** $\sec 2A + \csc 2A$

Question No. 19 (Only One Option Correct Type)

The value of

$$\int \frac{1}{[(x-1)^3(x+2)^5]} dx$$

is:

A. $\frac{4}{3} \left(\frac{x+1}{x-2} \right)^{\frac{1}{4}} + C$ **B.** $\frac{3}{4} \left(\frac{x-1}{x+2} \right)^{\frac{1}{4}} + C$ **C.** $\frac{4}{3} \left(\frac{x-1}{x+2} \right)^{\frac{1}{4}} + C$ **D.** $\frac{1}{3} \left(\frac{2x-1}{4x-3} \right)^{\frac{1}{4}} + C$

Question No. 20 (Only One Option Correct Type)

The number of solutions of the equation

$$\sin^{-1} x = (\sin x)^{-1}$$

is/are:

A. one B. two C. three D. zero

Mathematics Numerical (Maximum Marks: 20)

Question No. 1 (Numerical Type)

On the coordinate plane, ellipse $C_1 : \frac{x^2}{a_1^2} + \frac{y^2}{b_1^2} = 1$ (where $a_1 > b_1 > 0$) and hyperbola $C_2 : \frac{x^2}{a_2^2} - \frac{y^2}{b_2^2} = 1$ (where $a_2, b_2 > 0$) have the same focus points F_1, F_2 . Point P is one of the intersection points of C_1 and C_2 , and $PF_1 \perp PF_2$. If e_1 is the eccentricity of C_1 and e_2 is the eccentricity of C_2 , then find the minimum value of $9e_1^2 + e_2^2$.

Question No. 2 (Numerical Type)

If the equation

$$2x^2 + 4xy + 7y^2 - 12x - 2y + t = 0$$

where t is a parameter, has exactly one real solution of the form (x, y) , then the sum of $(x + y)$ is equal to:

Question No. 3 (Numerical Type)

If $f : [1, \infty) \rightarrow \mathbb{R}$ is a monotonic and differentiable function and $f(1) = 1$, then the number of solutions of the equation

$$f(f(x)) = \frac{1}{x^2 - 2x + 2}$$

is/are:

Question No. 4 (Numerical Type)

Let λ denote the number of terms in the expansion of

$$(1 + 5x + 10x^2 + 10x^3 + 5x^4 + x^5)^{20}.$$

If the unit's place and ten's place digits in 3^λ are O and T , respectively, then $O + T$ is equal to:

Question No. 5 (Numerical Type)

The area enclosed by the curve

$$2|x| + 3|y| = 6$$

on the xy -plane is:

(ANSWER KEY)

Physics

Q1 []	Q2 []	Q3 []	Q4 []	Q5 []
Q6 []	Q7 []	Q8 []	Q9 []	Q10 []
Q11[]	Q12 []	Q13 []	Q14 []	Q15 []
Q16[]	Q17 []	Q18 []	Q19 []	Q20 []

Q21 []	Q22 []	Q23 []	Q24 []	Q25 []
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Chemistry

Q1 []	Q2 []	Q3 []	Q4 []	Q5 []
Q6 []	Q7 []	Q8 []	Q9 []	Q10 []
Q11[]	Q12 []	Q13 []	Q14 []	Q15 []
Q16[]	Q17 []	Q18 []	Q19 []	Q20 []

Q21 []	Q22 []	Q23 []	Q24 []	Q25 []
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Mathematics

Q1 []	Q2 []	Q3 []	Q4 []	Q5 []
Q6 []	Q7 []	Q8 []	Q9 []	Q10 []
Q11[]	Q12 []	Q13 []	Q14 []	Q15 []
Q16[]	Q17 []	Q18 []	Q19 []	Q20 []

Q21 []	Q22 []	Q23 []	Q24 []	Q25 []
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